

Talgat Zharkynbay

“Module 1 Homework”

Load packages

```
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 3.5.3
```

```
library(nlme)  
library(car)
```

```
## Warning: package 'car' was built under R version 3.5.3
```

```
## Warning: package 'carData' was built under R version 3.5.2
```

Q1

i)

```
Signals=read.csv(file="Signals_A_62016.csv", header=TRUE, as.is = TRUE, na.strings = c("NA",  
".",""))
```

```
#Creating new dummy variables
```

```
Signals$ETP_IndNeg[is.na(Signals$ETP)] = 1  
Signals$ETP_IndNeg[!is.na(Signals$ETP)] = 0  
Signals$ETP_ValPos[is.na(Signals$ETP)] = 0  
Signals$ETP_ValPos[!is.na(Signals$ETP)] = Signals$ETP[!is.na(Signals$ETP)]  
#Creating Log Variables of Size and BTM
```

```
Signals = Signals %>%  
  mutate(LogSize = log(Size))  
Signals = Signals %>%  
  mutate(LogBTM = log(BTM))  
SignalsDF_1990 = Signals[Signals$PFormDate == "30/06/1990", ]
```

```
#Estimating Equation 1
```

```
EQ1=lm(AvgMonRet~LogSize+LogBTM+ETP_IndNeg+ETP_ValPos, data = SignalsDF_1990)  
summary(EQ1)
```

```
##
## Call:
## lm(formula = AvgMonRet ~ LogSize + LogBTM + ETP_IndNeg + ETP_ValPos,
##     data = SignalsDF_1990)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -23.498  -2.000  -0.108   1.632  49.821
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1.95132    0.28483   6.851 9.41e-12 ***
## LogSize       -0.13999    0.04403  -3.179  0.0015 **
## LogBTM         0.14844    0.10518   1.411  0.1583
## ETP_IndNeg     0.42397    0.25319   1.674  0.0942 .
## ETP_ValPos    -2.23473    1.64630  -1.357  0.1748
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.844 on 2285 degrees of freedom
## (7 observations deleted due to missingness)
## Multiple R-squared:  0.01327,    Adjusted R-squared:  0.01154
## F-statistic: 7.683 on 4 and 2285 DF,  p-value: 3.8e-06
```

We can see that a 1% increase in Size leads to a 0.0014 decrease in the Average Monthly Returns (AMR), while a 1% increase in Book to Market Ratio leads to approximately 0.0015 increase in the AMR. In addition, if a firm has a negative Earnings to Price Ratio then its AMR increases by 0.42 on average, while positive ETP decreases the AMR 2.23 on average. $R^2=0.01327$, while adjusted R^2 equals 0.01154. Although, only LogSize coefficient seems to be significant.

ii)

```

#Winsorizing Size
L = 0.005
H = 0.995
Size_L = quantile(Signals$Size, prob=L, na.rm=T)
Size_H = quantile(Signals$Size, prob=H, na.rm=T)
Signals$Size[Signals$Size<= Size_L] = Size_L
Signals$Size[Signals$Size>= Size_H] = Size_H

Signals = Signals %>%
  mutate(LogSize = log(Size))

#Winsorizing BTM
BTM_L = quantile(Signals$BTM, prob=L, na.rm=T)
BTM_H = quantile(Signals$BTM, prob=H, na.rm=T)
Signals$BTM[Signals$BTM<= BTM_L] = BTM_L
Signals$BTM[Signals$BTM>= BTM_H] = BTM_H

Signals = Signals %>%
  mutate(LogBTM = log(BTM))

#Winsorizing ETP_ValPos
ETP_ValPos_L = quantile(Signals$ETP_ValPos, prob=L, na.rm=T)
ETP_ValPos_H = quantile(Signals$ETP_ValPos, prob=H, na.rm=T)
Signals$ETP_ValPos[Signals$ETP_ValPos<= ETP_ValPos_L] = ETP_ValPos_L
Signals$ETP_ValPos[Signals$ETP_ValPos>= ETP_ValPos_H] = ETP_ValPos_H

SignalsDF_1990 = Signals[Signals$PFormDate == "30/06/1990", ]

EQ2=lm(AvgMonRet~LogSize+LogBTM+ETP_IndNeg+ETP_ValPos, data = SignalsDF_1990)
summary(EQ2)

```

```

##
## Call:
## lm(formula = AvgMonRet ~ LogSize + LogBTM + ETP_IndNeg + ETP_ValPos,
##     data = SignalsDF_1990)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -23.514  -2.002  -0.104   1.630   49.838
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1.95430     0.29114   6.712 2.41e-11 ***
## LogSize       -0.14200     0.04417  -3.215  0.00132 **
## LogBTM         0.13373     0.10841   1.233  0.21752
## ETP_IndNeg     0.42009     0.25757   1.631  0.10303
## ETP_ValPos    -2.26259     1.75863  -1.287  0.19838
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.844 on 2285 degrees of freedom
## (7 observations deleted due to missingness)
## Multiple R-squared:  0.01305,    Adjusted R-squared:  0.01132
## F-statistic: 7.551 on 4 and 2285 DF,  p-value: 4.85e-06

```

It looks like the data did not have much of outliers, since the coefficients have not changed significantly, and both the R^2 and adjusted R^2 have decreased.

iii)

```
#a)BTM 90th, ETP_ValPos 90th, Size 10th
```

```
SignalsDF_1990 = Signals[Signals$PFormDate == "30/06/1990", ]
```

```
LogSize3A=quantile(SignalsDF_1990$LogSize, prob=0.1, na.rm=T)
```

```
LogBTM3A=quantile(SignalsDF_1990$LogBTM, prob=0.9, na.rm=T)
```

```
ETP_ValPos3A=quantile(SignalsDF_1990$ETP_ValPos, prob=0.9, na.rm=T)
```

```
DF3A=data.frame(LogSize=LogSize3A, LogBTM=LogBTM3A, ETP_ValPos=ETP_ValPos3A,  
                ETP_IndNeg=0)
```

```
predict(EQ2, DF3A)
```

```
##      10%
```

```
## 1.37134
```

```
# Check how many such observations are in the sample:
```

```
SignalsDF_1990_3A=SignalsDF_1990 %>%
```

```
  filter(LogSize==quantile(Signals$LogSize, prob=0.1, na.rm=T),
```

```
         LogBTM==quantile(Signals$LogBTM, prob=0.9, na.rm=T),
```

```
         ETP_ValPos==quantile(Signals$ETP_ValPos, prob=0.9, na.rm=T))
```

```
summary(SignalsDF_1990_3A) # There are NO such observations!
```

```
##      PermNo      FDate      PFormDate      Size
## Min.      : NA    Length:0      Length:0      Min.      : NA
## 1st Qu.: NA    Class :character  Class :character  1st Qu.: NA
## Median : NA    Mode  :character  Mode  :character  Median : NA
## Mean   :NaN                      Mean   :NaN
## 3rd Qu.: NA                      3rd Qu.: NA
## Max.   : NA                      Max.   : NA
##      BTM      AG      IG      InvtG      NSI
## Min.      : NA    Min.      : NA    Min.      : NA    Min.      : NA    Min.      : NA
## 1st Qu.: NA    1st Qu.: NA    1st Qu.: NA    1st Qu.: NA    1st Qu.: NA
## Median : NA    Median : NA    Median : NA    Median : NA    Median : NA
## Mean   :NaN    Mean   :NaN    Mean   :NaN    Mean   :NaN    Mean   :NaN
## 3rd Qu.: NA    3rd Qu.: NA    3rd Qu.: NA    3rd Qu.: NA    3rd Qu.: NA
## Max.   : NA    Max.   : NA    Max.   : NA    Max.   : NA    Max.   : NA
##      NXF      OACC      ROE      ROA      ETP
## Min.      : NA    Min.      : NA    Min.      : NA    Min.      : NA    Min.      : NA
## 1st Qu.: NA    1st Qu.: NA    1st Qu.: NA    1st Qu.: NA    1st Qu.: NA
## Median : NA    Median : NA    Median : NA    Median : NA    Median : NA
## Mean   :NaN    Mean   :NaN    Mean   :NaN    Mean   :NaN    Mean   :NaN
## 3rd Qu.: NA    3rd Qu.: NA    3rd Qu.: NA    3rd Qu.: NA    3rd Qu.: NA
## Max.   : NA    Max.   : NA    Max.   : NA    Max.   : NA    Max.   : NA
##      AvgMonRet      CumAnnRet      Mkt_AvgMonRet      Mkt_CumMonRet      COMPNAME
## Min.      : NA    Min.      : NA    Min.      : NA    Min.      : NA    Length:0
## 1st Qu.: NA    1st Qu.: NA    1st Qu.: NA    1st Qu.: NA    Class :character
## Median : NA    Median : NA    Median : NA    Median : NA    Mode  :character
## Mean   :NaN    Mean   :NaN    Mean   :NaN    Mean   :NaN
## 3rd Qu.: NA    3rd Qu.: NA    3rd Qu.: NA    3rd Qu.: NA
## Max.   : NA    Max.   : NA    Max.   : NA    Max.   : NA
##      EXCHG      BBHL      FF48      ETP_IndNeg
## Min.      : NA    Length:0      Length:0      Min.      : NA
## 1st Qu.: NA    Class :character  Class :character  1st Qu.: NA
## Median : NA    Mode  :character  Mode  :character  Median : NA
## Mean   :NaN                      Mean   :NaN
## 3rd Qu.: NA                      3rd Qu.: NA
## Max.   : NA                      Max.   : NA
##      ETP_ValPos      LogSize      LogBTM
## Min.      : NA    Min.      : NA    Min.      : NA
## 1st Qu.: NA    1st Qu.: NA    1st Qu.: NA
## Median : NA    Median : NA    Median : NA
## Mean   :NaN    Mean   :NaN    Mean   :NaN
## 3rd Qu.: NA    3rd Qu.: NA    3rd Qu.: NA
## Max.   : NA    Max.   : NA    Max.   : NA
```

#b) BTM 10th, ETP 10th, Size 90th.

```
LogSize3B=quantile(SignalsDF_1990$LogSize, prob=0.9, na.rm=T)
```

```
LogBTM3B=quantile(SignalsDF_1990$LogBTM, prob=0.1, na.rm=T)
```

```
ETP_ValPos3B=quantile(SignalsDF_1990$ETP_ValPos, prob=0.1, na.rm=T)
```

```
DF3B=data.frame(LogSize=LogSize3B, LogBTM=LogBTM3B, ETP_ValPos=ETP_ValPos3B,
                ETP_IndNeg=0)
```

```
predict(EQ2, DF3B)
```

```
##      90%
## 0.6684057
```

```
# Check how many such observations are in the sample:
SignalsDF_1990_3B=SignalsDF_1990 %>%
  filter(LogSize==quantile(Signals$LogSize, prob=0.9, na.rm=T),
         LogBTM==quantile(Signals$LogBTM, prob=0.1, na.rm=T),
         ETP_ValPos==quantile(Signals$ETP_ValPos, prob=0.1, na.rm=T))

summary(SignalsDF_1990_3B)# There are NO such observations!
```

```
##      PermNo      FDate      PFormDate      Size
## Min.   : NA    Length:0      Length:0      Min.   : NA
## 1st Qu.: NA    Class :character Class :character 1st Qu.: NA
## Median : NA    Mode  :character Mode  :character Median : NA
## Mean   :NaN                                Mean   :NaN
## 3rd Qu.: NA                                3rd Qu.: NA
## Max.   : NA                                Max.   : NA
##      BTM      AG      IG      InvtG      NSI
## Min.   : NA    Min.   : NA    Min.   : NA    Min.   : NA    Min.   : NA
## 1st Qu.: NA    1st Qu.: NA    1st Qu.: NA    1st Qu.: NA    1st Qu.: NA
## Median : NA    Median : NA    Median : NA    Median : NA    Median : NA
## Mean   :NaN    Mean   :NaN    Mean   :NaN    Mean   :NaN    Mean   :NaN
## 3rd Qu.: NA    3rd Qu.: NA    3rd Qu.: NA    3rd Qu.: NA    3rd Qu.: NA
## Max.   : NA    Max.   : NA    Max.   : NA    Max.   : NA    Max.   : NA
##      NXF      OACC      ROE      ROA      ETP
## Min.   : NA    Min.   : NA    Min.   : NA    Min.   : NA    Min.   : NA
## 1st Qu.: NA    1st Qu.: NA    1st Qu.: NA    1st Qu.: NA    1st Qu.: NA
## Median : NA    Median : NA    Median : NA    Median : NA    Median : NA
## Mean   :NaN    Mean   :NaN    Mean   :NaN    Mean   :NaN    Mean   :NaN
## 3rd Qu.: NA    3rd Qu.: NA    3rd Qu.: NA    3rd Qu.: NA    3rd Qu.: NA
## Max.   : NA    Max.   : NA    Max.   : NA    Max.   : NA    Max.   : NA
##      AvgMonRet      CumAnnRet      Mkt_AvgMonRet      Mkt_CumMonRet      COMPNAME
## Min.   : NA    Min.   : NA    Min.   : NA    Min.   : NA    Length:0
## 1st Qu.: NA    1st Qu.: NA    1st Qu.: NA    1st Qu.: NA    Class :character
## Median : NA    Median : NA    Median : NA    Median : NA    Mode  :character
## Mean   :NaN    Mean   :NaN    Mean   :NaN    Mean   :NaN
## 3rd Qu.: NA    3rd Qu.: NA    3rd Qu.: NA    3rd Qu.: NA
## Max.   : NA    Max.   : NA    Max.   : NA    Max.   : NA
##      EXCHG      BBHL      FF48      ETP_IndNeg
## Min.   : NA    Length:0      Length:0      Min.   : NA
## 1st Qu.: NA    Class :character Class :character 1st Qu.: NA
## Median : NA    Mode  :character Mode  :character Median : NA
## Mean   :NaN                                Mean   :NaN
## 3rd Qu.: NA                                3rd Qu.: NA
## Max.   : NA                                Max.   : NA
##      ETP_ValPos      LogSize      LogBTM
## Min.   : NA    Min.   : NA    Min.   : NA
## 1st Qu.: NA    1st Qu.: NA    1st Qu.: NA
## Median : NA    Median : NA    Median : NA
## Mean   :NaN    Mean   :NaN    Mean   :NaN
## 3rd Qu.: NA    3rd Qu.: NA    3rd Qu.: NA
## Max.   : NA    Max.   : NA    Max.   : NA
```

```
#C) BTM 10th, ETP_Ind=0, Size 90th.
```

```
LogSize3C=quantile(SignalsDF_1990$LogSize, prob=0.9, na.rm=T)
```

```
LogBTM3C=quantile(SignalsDF_1990$LogBTM, prob=0.9, na.rm=T)
```

```
DF3C=data.frame(LogSize=LogSize3C, LogBTM=LogBTM3C, ETP_ValPos=0,  
                ETP_IndNeg=1)
```

```
predict(EQ2, DF3C)
```

```
##      90%
```

```
## 1.34051
```

```
# Check how many such observations are in the sample:
```

```
SignalsDF_1990_3C=SignalsDF_1990 %>%
```

```
  filter(LogSize==quantile(Signals$LogSize, prob=0.9, na.rm=T),
```

```
         LogBTM==quantile(Signals$LogBTM, prob=0.9, na.rm=T),
```

```
         ETP_ValPos==0, ETP_IndNeg==1)
```

```
# There are NO such observations!
```

To sum up, first strategy predicts the highest return of 1.37, hence our trading strategy should be holding a long position in firms with Small Size, and highest BTM and ETP ratios!

V)

```
#Fama-French 1992
```

```
#Winsorizing Size
```

```
L = 0.005
```

```
H = 0.995
```

```
Size_L = quantile(Signals$Size, prob=L, na.rm=T)
```

```
Size_H = quantile(Signals$Size, prob=H, na.rm=T)
```

```
Signals$Size[Signals$Size<= Size_L] = Size_L
```

```
Signals$Size[Signals$Size>= Size_H] = Size_H
```

```
Signals = Signals %>%
```

```
  mutate(LogSize = log(Size))
```

```
#Winsorizing BTM
```

```
BTM_L = quantile(Signals$BTM, prob=L, na.rm=T)
```

```
BTM_H = quantile(Signals$BTM, prob=H, na.rm=T)
```

```
Signals$BTM[Signals$BTM<= BTM_L] = BTM_L
```

```
Signals$BTM[Signals$BTM>= BTM_H] = BTM_H
```

```
Signals = Signals %>%
```

```
  mutate(LogBTM = log(BTM))
```

```
SignalsDF_1990 = Signals[Signals$PFormDate == "30/06/1990", ]
```

```
EQ3=lm(AvgMonRet~LogSize+LogBTM, data = SignalsDF_1990)
```

```
summary(EQ3)
```

```
##
## Call:
## lm(formula = AvgMonRet ~ LogSize + LogBTM, data = SignalsDF_1990)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -23.117  -2.004  -0.138   1.620  50.310
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   2.03340    0.21397   9.503  < 2e-16 ***
## LogSize      -0.17957    0.04174  -4.302 1.76e-05 ***
## LogBTM        0.06775    0.10311   0.657   0.511
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.85 on 2287 degrees of freedom
## (7 observations deleted due to missingness)
## Multiple R-squared:  0.009018,    Adjusted R-squared:  0.008151
## F-statistic: 10.41 on 2 and 2287 DF,  p-value: 3.173e-05
```

```
summary(EQ2)
```

```
##
## Call:
## lm(formula = AvgMonRet ~ LogSize + LogBTM + ETP_IndNeg + ETP_ValPos,
##     data = SignalsDF_1990)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -23.514  -2.002  -0.104   1.630  49.838
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1.95430    0.29114   6.712 2.41e-11 ***
## LogSize      -0.14200    0.04417  -3.215  0.00132 **
## LogBTM        0.13373    0.10841   1.233  0.21752
## ETP_IndNeg    0.42009    0.25757   1.631  0.10303
## ETP_ValPos   -2.26259    1.75863  -1.287  0.19838
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.844 on 2285 degrees of freedom
## (7 observations deleted due to missingness)
## Multiple R-squared:  0.01305,    Adjusted R-squared:  0.01132
## F-statistic: 7.551 on 4 and 2285 DF,  p-value: 4.85e-06
```

The adjusted R^2 has decreased from 0.01132 to 0.008151. Hence, we disagree with Fama and French.


```
#Sloan 1996
```

```
#Winsorizing OACC
```

```
OACC_L = quantile(Signals$OACC, prob=L, na.rm=T)
```

```
OACC_H = quantile(Signals$OACC, prob=H, na.rm=T)
```

```
Signals$OACC[Signals$OACC<= OACC_L] = OACC_L
```

```
Signals$OACC[Signals$OACC>= OACC_H] = OACC_H
```

```
SignalsDF_1990 = Signals[Signals$PFormDate == "30/06/1990", ]
```

```
EQ4=lm(AvgMonRet~LogSize+LogBTM+OACC, data = SignalsDF_1990)
```

```
summary(EQ2)
```

```
##
```

```
## Call:
```

```
## lm(formula = AvgMonRet ~ LogSize + LogBTM + ETP_IndNeg + ETP_ValPos,  
##     data = SignalsDF_1990)
```

```
##
```

```
## Residuals:
```

```
##      Min       1Q   Median       3Q      Max  
## -23.514  -2.002  -0.104   1.630  49.838
```

```
##
```

```
## Coefficients:
```

```
##              Estimate Std. Error t value Pr(>|t|)  
## (Intercept)  1.95430    0.29114   6.712 2.41e-11 ***  
## LogSize      -0.14200    0.04417  -3.215  0.00132 **  
## LogBTM        0.13373    0.10841   1.233  0.21752  
## ETP_IndNeg    0.42009    0.25757   1.631  0.10303  
## ETP_ValPos   -2.26259    1.75863  -1.287  0.19838
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
```

```
## Residual standard error: 3.844 on 2285 degrees of freedom
```

```
## (7 observations deleted due to missingness)
```

```
## Multiple R-squared:  0.01305,    Adjusted R-squared:  0.01132
```

```
## F-statistic: 7.551 on 4 and 2285 DF,  p-value: 4.85e-06
```

```
summary(EQ4)
```

```
##
## Call:
## lm(formula = AvgMonRet ~ LogSize + LogBTM + OACC, data = SignalsDF_1990)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -23.133  -1.973  -0.157   1.595  50.277
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   2.05300    0.21611   9.500 < 2e-16 ***
## LogSize      -0.17709    0.04203  -4.213 2.62e-05 ***
## LogBTM        0.07396    0.10855   0.681  0.496
## OACC         -0.03001    0.84720  -0.035  0.972
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.807 on 2185 degrees of freedom
## (108 observations deleted due to missingness)
## Multiple R-squared:  0.009269, Adjusted R-squared:  0.007909
## F-statistic: 6.814 on 3 and 2185 DF, p-value: 0.0001435
```

Again adjusted R^2 has decreased, and so did the explanatory power, hence we disagree with Sloan too.

```
#VIF
vif(EQ4)
```

```
## LogSize LogBTM OACC
## 1.050066 1.053168 1.003660
```

VIF of LogSize on LogBTM is 1.050066, and VIF of OACC is 1.003660. Since all values are very close to 1, it means they are almost perfectly uncorrelated.

vi)

```
#We have already winsorized the data in the previous task.
#We start by creating new dataset.

SignalsDF_New=Signals %>%
  filter(LogSize!="NA",LogBTM!="NA",AvgMonRet!="NA",
         PFormDate>="30/06/1990",PFormDate<="30/06/2015")

SignalsDF_New=SignalsDF_New %>%
  select("PFormDate", "LogSize", "LogBTM", "AvgMonRet")

#Then we do regressions

fit_FM = lmList(AvgMonRet ~ LogSize+LogBTM| factor(PFormDate), data=SignalsDF_New)

summary_fit_FM = summary(fit_FM)

summary(coefficients(summary_fit_FM)[,2])#This is estimates of LogSize or B1
```

```
##      Estimate      Std. Error      t value      Pr(>|t|)
## Min.      :-1.0966   Min.      :0.04955   Min.      :-21.648   Min.      :0.000e+00
## 1st Qu.: -0.4364   1st Qu.:0.05212   1st Qu.:  -7.959   1st Qu.:0.000e+00
## Median : -0.2262   Median :0.05339   Median :  -4.324   Median :1.601e-05
## Mean      :-0.2856   Mean      :0.05322   Mean      : -5.404   Mean      :1.842e-02
## 3rd Qu.: -0.1189   3rd Qu.:0.05445   3rd Qu.:  -2.204   3rd Qu.:2.130e-02
## Max.       : 0.1279   Max.       :0.05554   Max.       :  2.423   Max.       :2.233e-01
```

```
summary(coefficients(summary_fit_FM)[,3])#This is estimates of LogBTM or B2
```

```
##      Estimate      Std. Error      t value      Pr(>|t|)
## Min.      :-1.5392   Min.      :0.09363   Min.      :-14.289   Min.      :0.0000000
## 1st Qu.: -0.1872   1st Qu.:0.11315   1st Qu.:  -1.397   1st Qu.:0.0000000
## Median :  0.1962   Median :0.12124   Median :   1.561   Median :0.0008332
## Mean      : 0.2025   Mean      :0.12042   Mean      :   1.851   Mean      :0.1055394
## 3rd Qu.:  0.6266   3rd Qu.:0.12726   3rd Qu.:   5.351   3rd Qu.:0.1368119
## Max.       : 1.5043   Max.       :0.13849   Max.       : 14.866   Max.       :0.7162692
```

```
summary(coefficients(fit_FM)) # Table of estimates
```

```
##      (Intercept)      LogSize      LogBTM
## Min.      :-1.690   Min.      :-1.0966   Min.      :-1.5392
## 1st Qu.:  2.192   1st Qu.: -0.4364   1st Qu.: -0.1872
## Median :  3.491   Median : -0.2262   Median :  0.1962
## Mean      :  3.425   Mean      :-0.2856   Mean      : 0.2025
## 3rd Qu.:  4.305   3rd Qu.: -0.1189   3rd Qu.:  0.6266
## Max.       :  8.327   Max.       : 0.1279   Max.       : 1.5043
```

```
summary((summary_fit_FM)$r.squared) #This is R-squared
```

```
##      Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
## 0.001297 0.008583 0.025443 0.031872 0.044117 0.112767
```

According to results, on average LogBTM has a positive effect on returns, while LogSize has negative effect. R-squared equals 0.031872 on average.

vii)

```
#Let's reupload the Signals data to avoid possible mistakes:
Signals=read.csv(file="Signals_A_62016.csv", header=TRUE, as.is = TRUE, na.strings = c("NA",
".", ""))

Signals = Signals %>%
  mutate(LogSize = log(Size))
Signals = Signals %>%
  mutate(LogBTM = log(BTM))
SignalsDF_1990 = Signals[Signals$PFormDate == "30/06/1990", ]

SignalsDF_1990Q7 <- SignalsDF_1990 %>% #NEW Column by classification
  mutate(EXCHNG_11 = ifelse(EXCHG ==11, 1, 0),
    EXCHNG_12 = ifelse(EXCHG ==12, 1, 0),
    EXCHNG_14 = ifelse(EXCHG ==14, 1, 0))

#Omit LogBTM*EXCNG_11 (c) Doron

EQ5=lm(AvgMonRet ~ 0+LogSize+LogBTM+EXCHNG_11+EXCHNG_12+EXCHNG_14
  +LogBTM:EXCHNG_12+LogBTM:EXCHNG_14,
  data = SignalsDF_1990Q7)

summary(EQ5)
```

```
##
## Call:
## lm(formula = AvgMonRet ~ 0 + LogSize + LogBTM + EXCHNG_11 + EXCHNG_12 +
##     EXCHNG_14 + LogBTM:EXCHNG_12 + LogBTM:EXCHNG_14, data = SignalsDF_1990Q7)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -23.062  -1.972  -0.150   1.685   50.129
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## LogSize          -0.27630     0.05212  -5.301 1.26e-07 ***
## LogBTM             0.15634     0.16273   0.961 0.336799
## EXCHNG_11          2.86123     0.33487   8.544 < 2e-16 ***
## EXCHNG_12          1.16844     0.33070   3.533 0.000419 ***
## EXCHNG_14          2.42044     0.23805  10.168 < 2e-16 ***
## LogBTM:EXCHNG_12  -0.35180     0.35185  -1.000 0.317481
## LogBTM:EXCHNG_14  -0.09774     0.21142  -0.462 0.643910
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.833 on 2283 degrees of freedom
## (7 observations deleted due to missingness)
## Multiple R-squared:  0.09659,    Adjusted R-squared:  0.09382
## F-statistic: 34.87 on 7 and 2283 DF,  p-value: < 2.2e-16
```

viii)

Regression without intercept is called the regression through origin, This model omits b_0 , because it would not make sense for a firm to have Returns if its Size was zero. If we estimated a model with the intercept we would have unbiased OLS estimators and ensure Zero conditional mean, (MLR3-4). This model certainly needs to

include the intercept so follow Gauss-Markov assumptions. * * *

Q2

i)

b0 is the intercept, b1 is the AEBE Effect of college on salary (log-level relationship), b2 estimates how percentage increase in sales leads to a percentage increase in salary. b3 measures the percentage effect of profits, b4 measures the effect of tenure, and b5 measures the effect of squared tenure. So that the effect of tenure becomes $b4 + b5 \cdot \text{CeoTenure}$!

ii)

```
CEOSAL2=read.table(file="CEOSAL2.txt", sep="", header=FALSE, as.is = TRUE, na.strings = c("N
A",".", ""))
CEOSAL2=CEOSAL2[,c('V1','V3','V6','V7','V8','V14')]
colnames(CEOSAL2)=c("salary","college", "ceoten", "sales", "profits","ceotensq")

CEOSAL2$profits[CEOSAL2$profits<0]=NA
CEOSAL2_NEW=na.omit(CEOSAL2)
fit1_p2=lm(log(salary)~college+log(sales)+log(profits)+ceoten+ceotensq,data=CEOSAL2_NEW)
summaryfit1_p2=summary(fit1_p2)
summaryfit1_p2
```

```
##
## Call:
## lm(formula = log(salary) ~ college + log(sales) + log(profits) +
##      ceoten + ceotensq, data = CEOSAL2_NEW)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.27530 -0.29490 -0.01555  0.30620  1.70847
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  4.6159114  0.3295736  14.006  < 2e-16 ***
## college      0.0083331  0.2303410   0.036  0.97119
## log(sales)   0.2273347  0.0482829   4.708 5.33e-06 ***
## log(profits) 0.0229473  0.0529849   0.433  0.66552
## ceoten       0.0435020  0.0145906   2.982  0.00331 **
## ceotensq     -0.0011955  0.0004848  -2.466  0.01470 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5006 on 162 degrees of freedom
## Multiple R-squared:  0.3523, Adjusted R-squared:  0.3323
## F-statistic: 17.62 on 5 and 162 DF,  p-value: 6.404e-14
```

If a CEO attended a college, the salary is expected to increase by 0.83331%, increase in sales by 1% increases salary by 0.227%, while profits increase salary only by 0.023%. The Tenure seems to have a parabolic shape, since it increases salary in the beginning, but it tends to decrease the salary if a tenure becomes too large.

iii)

```
newDF_p2=data.frame(college=0,sales=quantile(CEOSAL2_NEW$sales, prob=0.5, na.rm=T),profits=quantile(CEOSAL2_NEW$profits, prob=0.7, na.rm=T),ceoten=10,ceotensq=100)
predict(fit1_p2,newDF_p2)
```

```
##      50%
## 6.697666
```

iv)

```
fit2_p2=lm(salary~college+sales+profits+ceoten+ceotensq,data=CEOSAL2_NEW)
summaryfit2_p2=summary(fit2_p2)
summaryfit2_p2
```

```
##
## Call:
## lm(formula = salary ~ college + sales + profits + ceoten + ceotensq,
##     data = CEOSAL2_NEW)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -959.9  -294.8   -66.4   263.3  4331.0
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  584.26294   257.91484    2.265  0.02482 *
## college      -99.66103   241.10518   -0.413  0.67990
## sales         0.02001     0.01175    1.703  0.09050 .
## profits       0.35673     0.18033    1.978  0.04960 *
## ceoten       46.39950    15.31930    3.029  0.00286 **
## ceotensq     -1.24429     0.50924   -2.443  0.01562 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 526.4 on 162 degrees of freedom
## Multiple R-squared:  0.231, Adjusted R-squared:  0.2073
## F-statistic: 9.734 on 5 and 162 DF, p-value: 3.8e-08
```

```
MoM_Est=mean(exp(fit1_p2$residuals));MoM_Est
```

```
## [1] 1.129431
```

```
MoM_FitVal=MoM_Est*exp(fit1_p2$fitted.values)
(cor(CEOSAL2_NEW$salary,MoM_FitVal))^2
```

```
## [1] 0.2900913
```

```
summaryfit2_p2$r.squared
```

```
## [1] 0.2310196
```

We would disagree, because the Pseudo R^2 of a new model is equal to 0.23, it is less than 0.3323 Adjusted R^2 for the first model.